Final Report:
Mach-Zehnder Interferometer

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1. Introduction

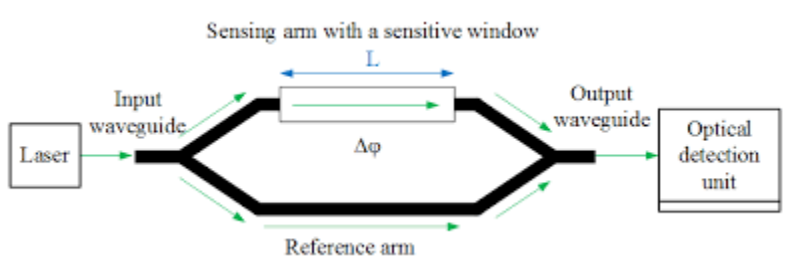
Silicon photonics has established itself as a pivotal technology for high-density integrated optics, driven largely by its compatibility with the mature complementary metal-oxide-semiconductor (CMOS) manufacturing infrastructure (Bogaerts & Chrostowski, 2018). This platform typically utilizes Silicon-on-Insulator (SOI) wafers, where the high refractive index contrast between the silicon core and the silicon dioxide cladding enables the tight confinement of light within sub-micron waveguides (Bryant et al., 2025). This confinement capability allows for the creation of compact photonic integrated circuits (PICs) with small bend radii, essential for scaling optical networks and computing systems.

A fundamental component within these circuits is the Mach-Zehnder Interferometer (MZI). The MZI serves as a versatile building block for manipulating the phase and amplitude of optical signals (Sun et al., 2022). Its operational principle—based on the interference of split optical paths—has enabled a wide array of applications, ranging from high-speed electro-optic modulation and optical switching to highly sensitive environmental sensing (Sultan et al., 2023; Sun et al., 2022). This report explores the design and theoretical underpinnings of MZI devices based on strip silicon waveguides, examining how geometric parameters and path asymmetries influence spectral performance.

2. Theory

2.1 Operating Principle

The operation of an integrated MZI relies on the splitting and subsequent recombination of a coherent optical signal. Light entering the device is divided into two separate arms, typically via a directional coupler or a multimode interference (MMI) splitter. As the light propagates through these arms, it accumulates a phase shift ϕ determined by the waveguide's effective refractive index n_{eff} and the physical length L of the path (Sun et al., 2022).



In an unbalanced MZI configuration, a physical path length difference ΔL is introduced between the two arms. The resulting phase difference $\Delta\phi$ between the propagating signals is wavelength-dependent and is given by:

$$\Delta\phi(\lambda) = \frac{2\pi n_{eff}(\lambda)\Delta L}{\lambda}$$

where λ is the operating wavelength in vacuum (Sun et al., 2022).

2.2 Transmission Characteristics

When the optical signals from the two arms recombine, they interfere either constructively or destructively depending on their relative phase difference. For an ideal lossless MZI with a 50:50 splitting ratio, the normalized output intensity I_{out} follows a sinusoidal transfer function described by:

$$I_{out} = \frac{1}{2}[1 + \cos(\Delta\phi(\lambda))] \quad (1)$$

This interference mechanism results in a periodic spectral response characterized by transmission maxima (constructive interference) and minima (destructive interference) (Sun et al., 2022).

2.3 Free Spectral Range and Group Index

A critical performance metric for interferometric devices is the Free Spectral Range (FSR), defined as the wavelength spacing between consecutive transmission peaks. The FSR is inversely proportional to the path length imbalance and the group index n_g of the waveguide (Bryant et al., 2025). The relationship is expressed as:

$$FSR \approx \frac{\lambda^2}{n_g \Delta L}$$

The group index n_g accounts for the dispersive nature of the waveguide medium, incorporating both material and waveguide dispersion effects. It is derived from the effective index using the following relation (Dulkeith et al., 2006):

$$n_g(\lambda) = n_{eff}(\lambda) - \lambda\frac{dn_{eff}}{d\lambda}$$

Accurate modeling of n_{eff} and n_g is essential for predicting the MZI's spectral behavior, particularly for applications requiring precise wavelength filtering or modulation (Bryant et al., 2025).

3. Modeling and Simulations:

3.1 Waveguide Modeling:

The foundational component for the Mach-Zehnder Interferometer (MZI) is the silicon-on-insulator (SOI) strip waveguide. We considered a standard low-loss silicon rectangle embedded within a silicon dioxide layer. The vertical dimension, or height, of the silicon core was fixed at 220 nm, which is a common thickness for high-volume silicon photonics manufacturing. The waveguide width was intentionally set to 500 nm. This specific width was chosen to optimize for single-mode operation while maintaining low propagation loss. Operating in the single-mode regime is critical to prevent unwanted coupling and ensure predictable device performance. This selection represents a necessary compromise: narrower waveguides experience greater optical field interaction with the rough sidewalls, leading to increased sidewall scattering and loss, whereas wider waveguides support multiple optical modes, which would compromise the MZI's single-mode operation.

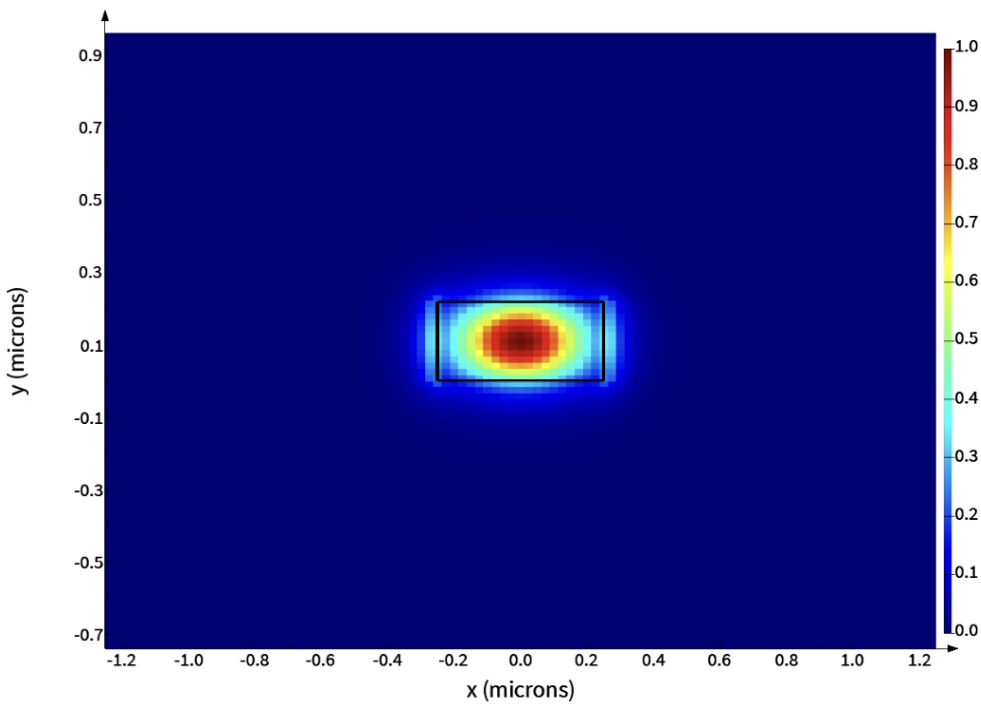


Figure 2: $|E|^2$ mode for a 220 x 500 nm SOI waveguide.

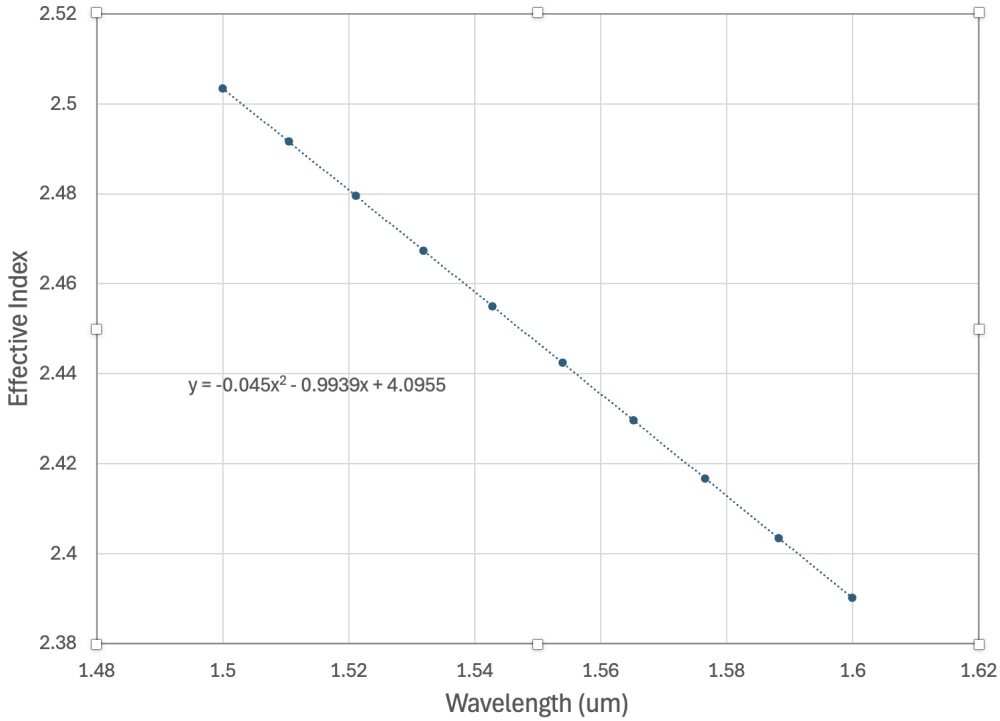


Figure 3: Effective index for a 220 x 500 nm SOI waveguide vs wavelength.

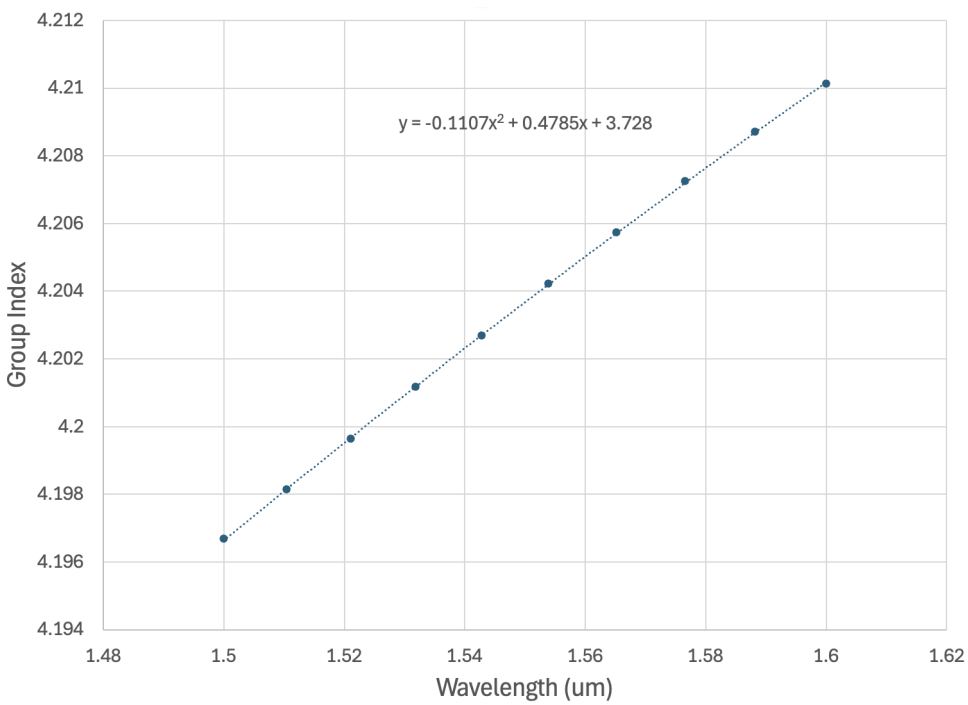


Figure 4: Group index for a 220 x 500 nm SOI waveguide vs wavelength.

The optical characteristics of this 220 nm x 500 nm SOI waveguide were thoroughly simulated. Figure 2 illustrates the calculated $|E|^2$ mode profile, demonstrating the tight light confinement. Figures 3 and 4 present the calculated effective index and group index as a function of wavelength, respectively. Based on these simulation results, polynomial fits were extracted to provide accurate, analytical expressions for the indices across the operating wavelength range. These expressions were used in subsequent MZI transmission models:

$$n_{eff} = -0.045\lambda^2 - 0.9939\lambda + 4.0955$$

$$n_g = -0.1107\lambda^2 + 0.4785\lambda + 3.728$$

3.2 Mach-Zehnder Interferometer Modeling:

The MZI circuit model was constructed using the Lumerical Interconnect simulation tool. As shown in Figure 5, the model incorporated key passive components essential to the device structure: an input grating coupler, a y-splitter functioning as the 50:50 power divider, two waveguide arms to introduce the phase difference, a second y-splitter for signal recombination, and an output grating coupler.

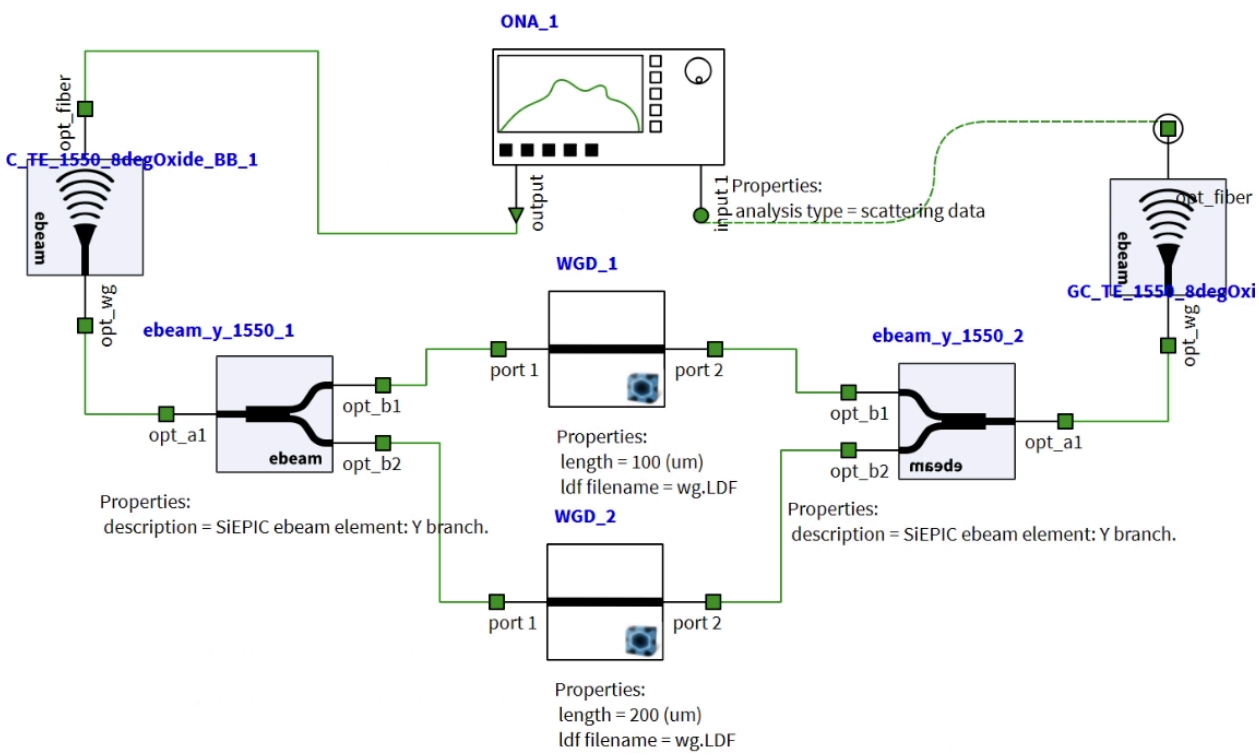


Figure 5: Lumerical Interconnect model for our MZI.

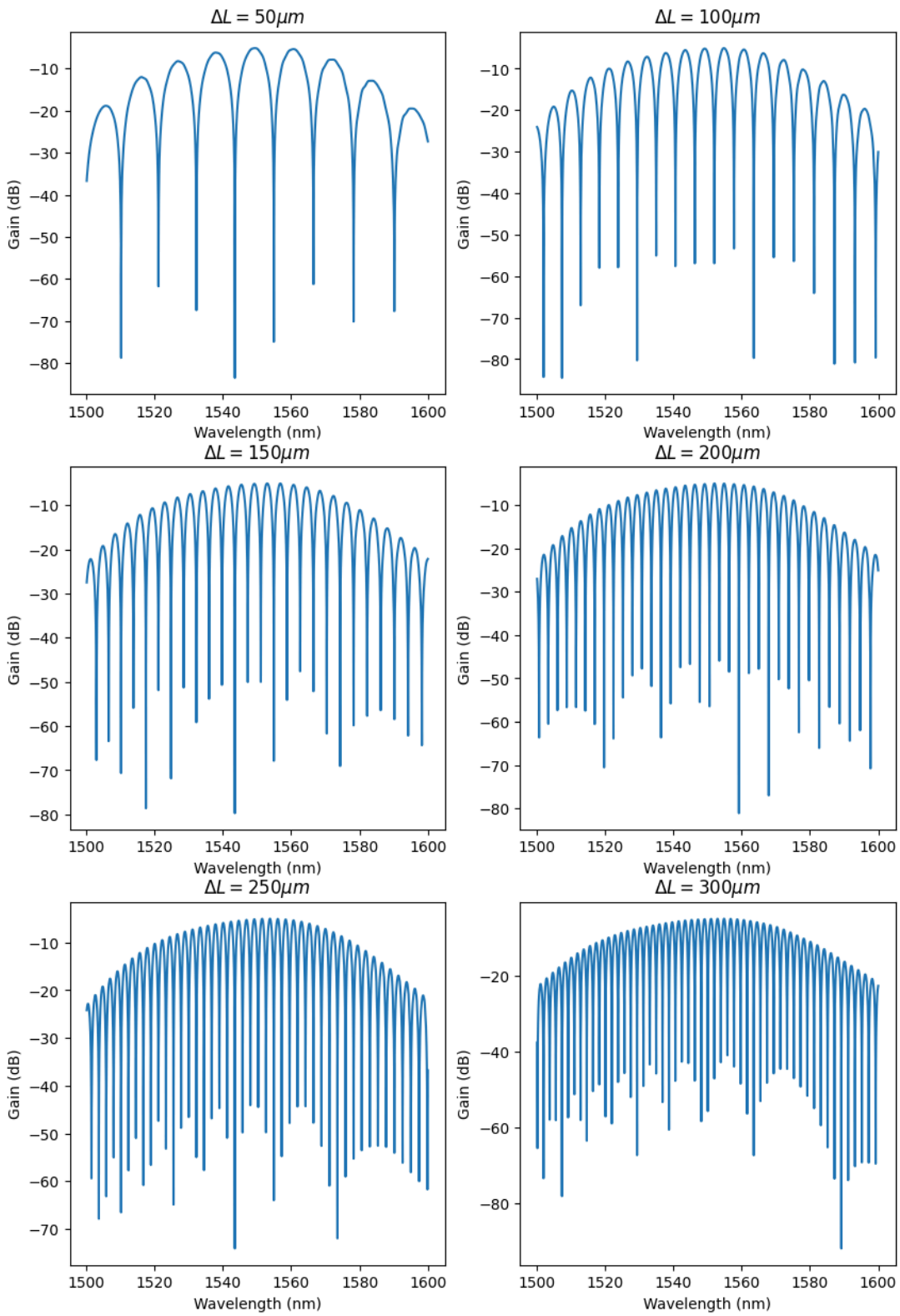


Figure 6: MZI response function vs wavelength for a variety of imbalanced lengths $\Delta L = 50, 100, 150, 200, 250, 300\ \mu m$.

The simulation specifically accounted for wavelength-dependent losses present in both the grating couplers and the waveguides. By varying the physical length difference (ΔL) between the two arms, we simulated the spectral response of the MZI. The simulated results, presented in Figure 6 for path length differences ranging from 50 to 300 µm, confirmed the inverse relationship between the path length difference and the Free Spectral Range (FSR). Specifically, increasing the path length difference resulted in a decrease in the FSR, as predicted by the theoretical model.

3.3 Corner Analysis of Fabrication Errors:

To estimate the boundary of possible performance variability due to manufacturing tolerances, a corner analysis was conducted. This approach defines the worst-case parameter space by considering the extreme limits of expected fabrication variation, providing a deterministic estimate of performance deviation. The analysis focused on two primary fabrication parameters: the SOI wafer thickness and the waveguide width. We assumed the wafer thickness varied between 215.3 nm and 223.1 nm, and the waveguide width varied between 470 nm and 510 nm. The nominal design (220 nm thickness and 500 nm width) and the edge cases defined by these maximum and minimum tolerances were analyzed. This analysis was conducted for a path length difference of 50 µm.
The results of the corner analysis, visualized in Figure 7, indicated the following range of variability for device characteristics at a wavelength of 1550 nm:

- Effective index: 2.37 - 2.48.
- Group index: 4.18 - 4.263.
- FSR: 11.3 - 11.5 nm.

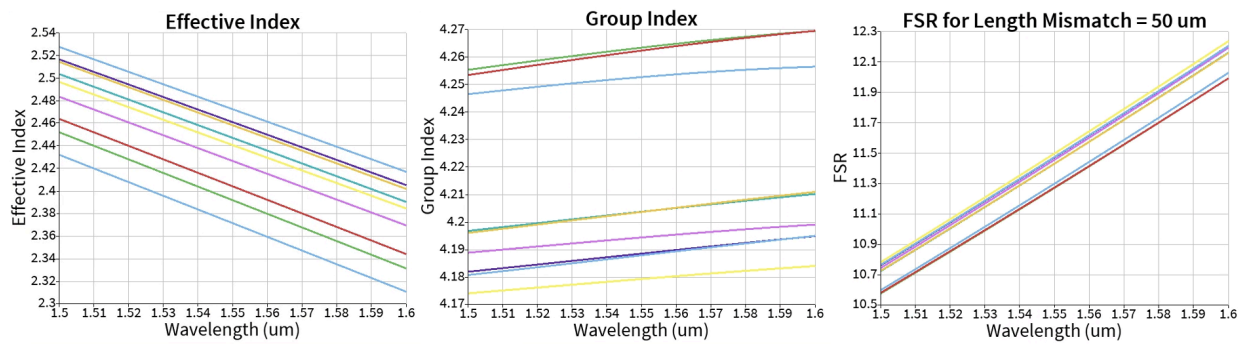


Figure 7: Corner analysis of effective index, group index, and FSR ($\Delta L = 50\ \mu m$) in a MZI

3.4 Monte-Carlo Analysis of Fabrication Errors:

In contrast to the corner analysis, Monte-Carlo simulations were performed to provide a statistical prediction of fabrication variability. This method involves sampling a large number of devices from a probability distribution that accurately represents the expected manufacturing process variation. For our analysis, we modeled the waveguide width and thickness variations using a Gaussian distribution, specifying a standard deviation of 5 nm for the width and 2 nm for the thickness. We performed 100 separate simulations based on these statistical variations

The transmission results of the 100 simulated devices (Figure 8) show that the random variation in waveguide dimensions causes a spectral shift, resulting in the peaks "wandering" across the spectral range from one device simulation to the next. This effect highlights the sensitivity of the MZI's spectral position to manufacturing tolerances. Figure 9 presents a histogram of the Free Spectral Range (FSR) values calculated from the 100 simulations, which provides a statistical distribution of this key performance metric for a path length mismatch of $\Delta L = 50\ \mu m$..

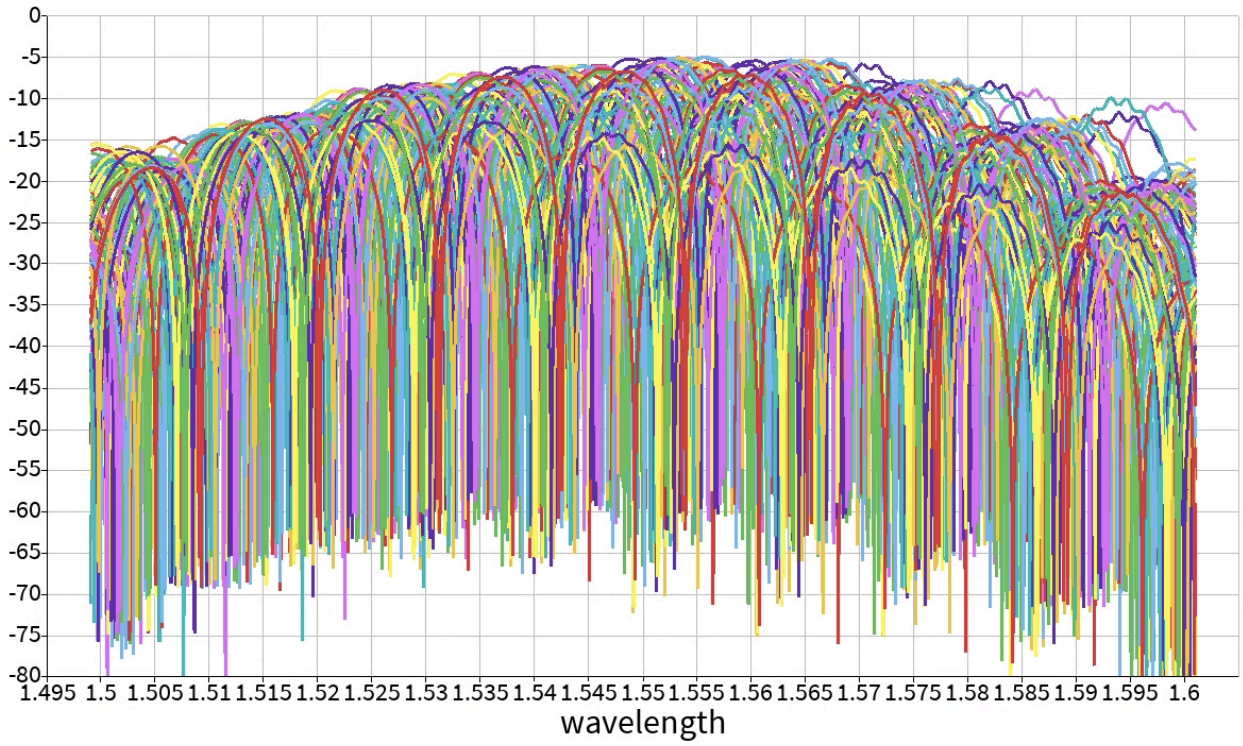


Figure 8: Monte-Carlo analysis of 100 devices displaying the spectra of an MZI of length mismatch $\Delta L = 50\ \mu m$

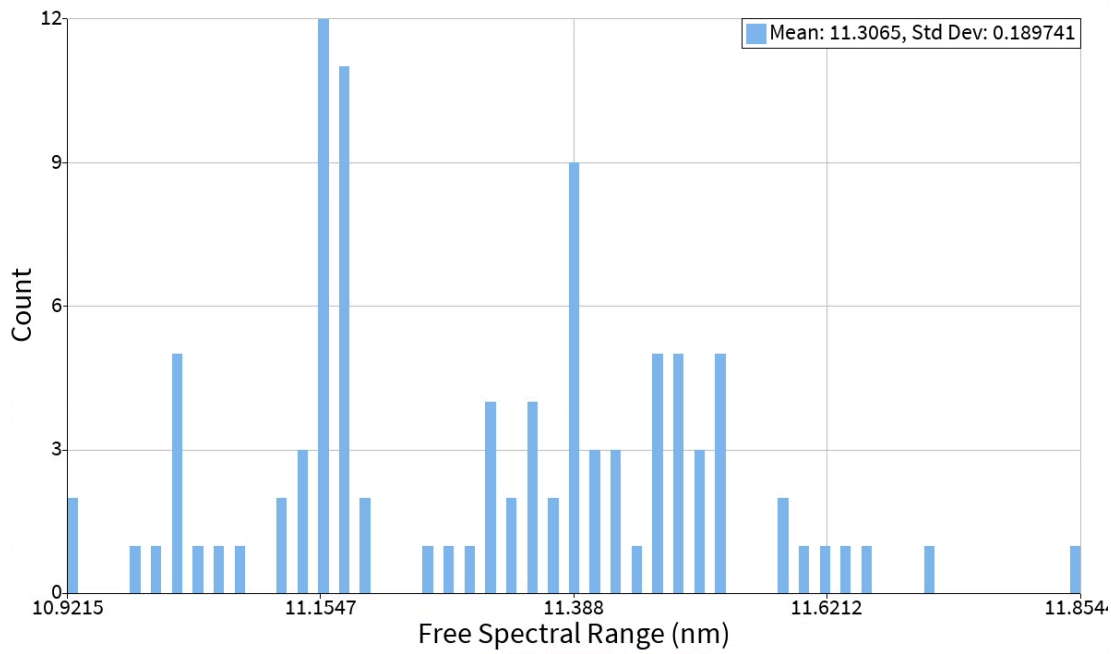


Figure 9: Monte-Carlo analysis of 100 devices displaying the free spectral range of an MZI of length mismatch $\Delta L = 50\ \mu m$

4. Experimental Results:

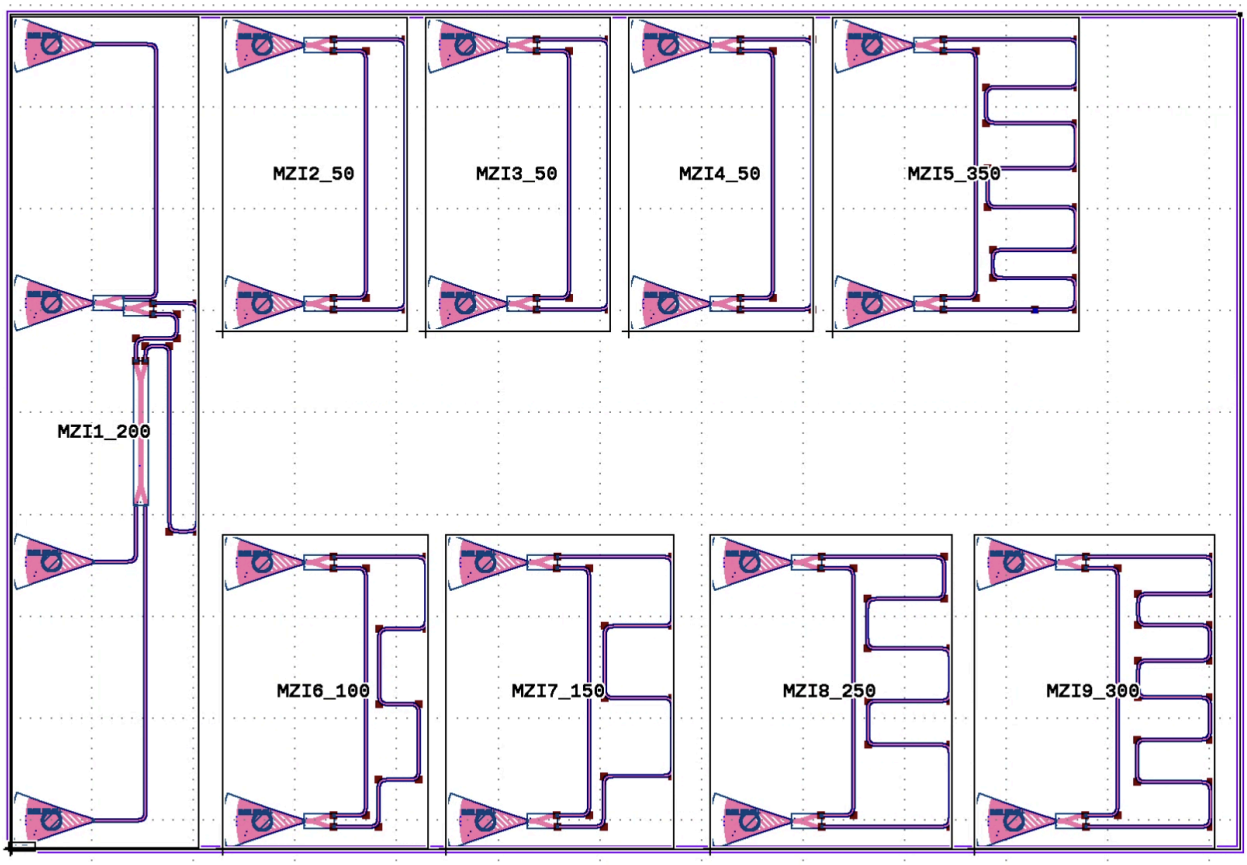


Figure 10: GDS Layout for the fabricated design that consists of 9 devices of different path length differences.

Table 1: Path length difference for each fabricated Mach-Zehnder Interferometer in our design.

Name	ΔL (μm)
MZI1	200
MZI2	50
MZI3	50
MZI4	50
MZI5	350
MZI6	100
MZI7	150

MZI8	250
MZI9	300

The photonic devices were fabricated using the NanoSOI MPW fabrication process by Applied Nanotools Inc. (http://www.appliednt.com/nanosoi; Edmonton, Canada) which is based on direct-write 100 keV electron beam lithography technology. Silicon-on-insulator wafers of 200 mm diameter, 220 nm device thickness and 2 μm buffer oxide thickness are used as the base material for the fabrication. The wafer was pre-diced into square substrates with dimensions of 25x25 mm, and lines were scribed into the substrate backsides to facilitate easy separation into smaller chips once fabrication was complete. After an initial wafer clean using piranha solution (3:1 H2SO4:H2O2) for 15 minutes and water/IPA rinse, hydrogen silsesquioxane (HSQ) resist was spin-coated onto the substrate and heated to evaporate the solvent. The photonic devices were patterned using a JEOL JBX-8100FS electron beam instrument at The University of British Columbia. The exposure dosage of the design was corrected for proximity effects that result from the backscatter of electrons from exposure of nearby features. Shape writing order was optimized for efficient patterning and minimal beam drift. After the e-beam exposure and subsequent development with a tetramethylammonium sulfate (TMAH) solution, the devices were inspected optically for residues and/or defects. The chips were then mounted on a 4” handle wafer and underwent an anisotropic ICP-RIE etch process using chlorine after qualification of the etch rate. The resist was removed from the surface of the devices using a 10:1 buffer oxide wet etch, and the devices were inspected using a scanning electron microscope (SEM) to verify patterning and etch quality. A 2.2 μm oxide cladding was deposited using a plasma-enhanced chemical vapour deposition (PECVD) process based on tetraethyl orthosilicate (TEOS) at 300ºC. Reflectrometry measurements were performed throughout the process to verify the device layer, buffer oxide and cladding thicknesses before delivery.

Figure 10 displays the GDS layout for our field within the fabricated wafer. The layout includes 9 devices each of which is a Mach-Zehnder interferometer. Table 1 details the path length differences for each interferometer. We chose to conduct the following experiments:

1. A loopback in MZI 1 to normalize our interferometer spectrum.
2. A 2 output interferometer in MZI 1 to display the phase difference between the two outputs.
3. 3 identical interferometers in MZI 2, 3, and 4 to study fabrication variation.
4. Interferometers of different path length differences ranging from 50 - 350 μm.

We detail the results of these experiments below.

4.1 Loopback and Dual-Output Interferometer:

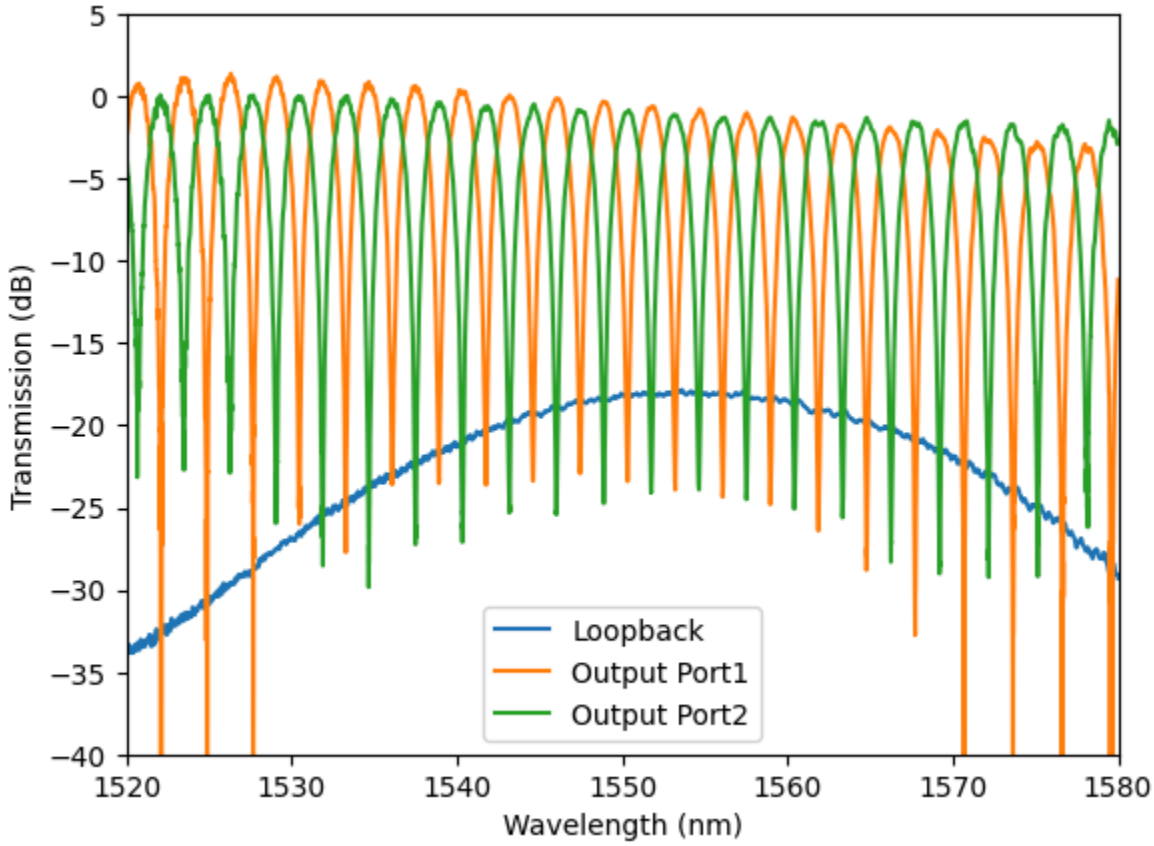


Figure 11:Transfer function of MZI 1 ($\Delta L = 200\ \mu m$) across three output ports, (i) without interferometer (loopback) and (ii) with interferometer on both output ports. The interferometer results are normalized by removing the loopback spectrum.

The initial step in characterizing the Mach-Zehnder Interferometer (MZI) is to accurately determine the spectral response of the fundamental passive components: the input/output gratings and the connecting waveguide. This is achieved by first measuring a dedicated "loopback" structure, which consists of a defined length of waveguide flanked by a pair of input and output coupling gratings. The resulting spectrum from this loopback structure, represented by the blue curve in Figure 11, encapsulates the combined spectral roll-off and insertion loss characteristics inherent to these components. This measured loopback spectrum is critical because it represents the parasitic non-interferometric response that must be decoupled from the MZI's intended functionality. Therefore, this loopback spectrum is subsequently subtracted from all measured MZI spectra to isolate and retrieve the pure interferometric response, allowing for a precise analysis of the interference mechanism itself.

Furthermore, MZI 1 was deliberately engineered with a dual-port output configuration. This design incorporates two distinct output ports, each directed to its own dedicated outcoupling grating. The normalized optical power spectra collected from these two separate output ports are depicted in Figure 11 by the orange and green curves, respectively. The analysis of these spectra provides crucial validation for the device's performance. Specifically, the Free Spectral Range (FSR) extracted from the periodic oscillations of both curves exhibits excellent agreement with theoretical predictions based on the designed path length difference within the interferometer's arms. Crucially, the fundamental principle of energy conservation mandates a specific relationship between the two output ports. As anticipated and confirmed by the measurements, the two output spectra are intrinsically out-of-phase with one another. This relationship is manifest in the spectral domain where the constructive interference maxima observed at one port correspond precisely to the destructive interference minima at the other port, demonstrating the expected power transfer between the two outputs.

4.2 Repeatability Study:

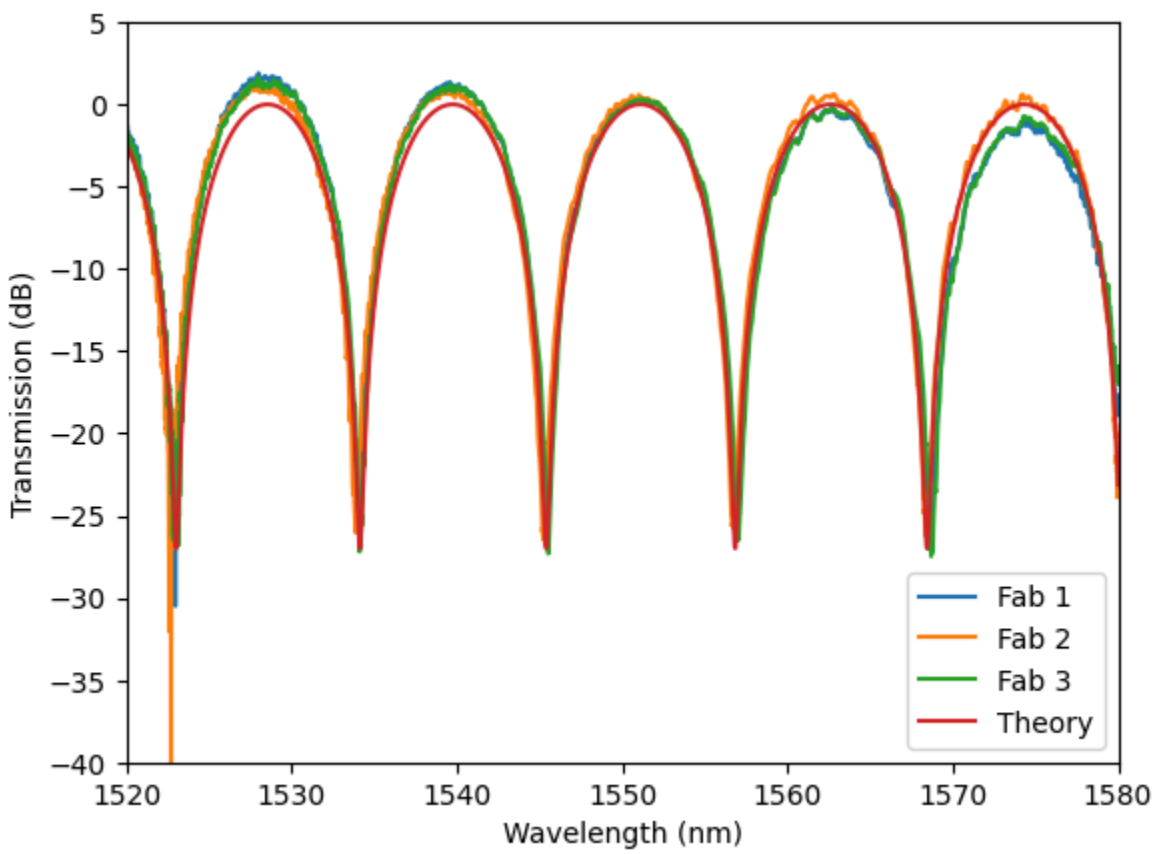


Figure 12:Transfer function of MZI 2-4 ($\Delta L = 50\ \mu m$) as compared to the theoretical transfer function for a Mach-Zehnder interferometer with $n_{eff}(\lambda = 1550\ nm) = 2.452$ and $n_g(\lambda = 1550\ nm) = 4.2087$.

This experimental investigation centered on the fabrication and characterization of three identical Mach-Zehnder Interferometers (MZIs). A key design parameter for these devices was the substantial path length difference, meticulously controlled to $\Delta L = 50\ \mu m$, which dictates the free spectral range (FSR) and overall spectral response of the interferometers.

Figure 12 provides a crucial visual representation of the experimental outcome, displaying the normalized measured optical power spectra for each of the three fabricated MZI devices. This empirical data is then directly overlaid and compared against the theoretical transfer function, which is mathematically described by Equation 1.

The results demonstrate a high level of performance and consistency across the fabricated batch. Specifically:

1. **Repeatability:** The measured spectra from all three separate interferometers show near-perfect congruence with one another. This high degree of overlap confirms the robustness and precision of the fabrication process, indicating minimal device-to-device variation and predictable performance.
2. **Near-Perfect Agreement with Theory:**, the normalized empirical spectra align almost flawlessly with the shape, period, and intensity predicted by the theoretical transfer function (Eq. 1). This strong correlation validates the fundamental design principles and the theoretical model used to predict the MZI behavior based on the established path length difference.

In summary, the successful realization of three highly repeatable interferometers, whose measured spectral characteristics closely mirror the theoretical predictions, confirms both the quality of the device fabrication and the accuracy of the underlying optical model.

4.3 Interferometers with different path length differences:

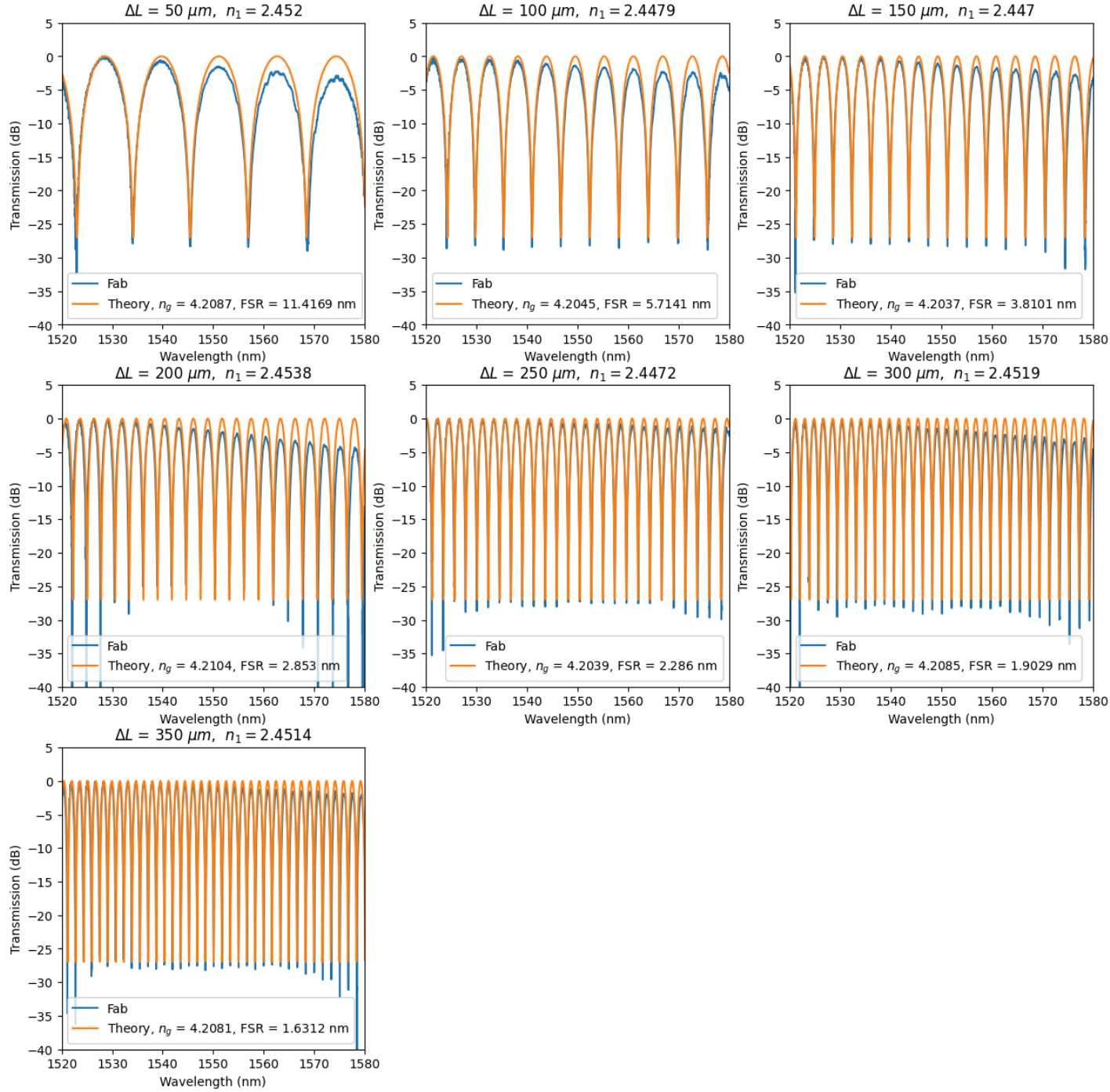


Figure 13:Transfer function of all our interferometers as compared to the theoretical transfer function for a Mach-Zehnder interferometer. For each case the effective index, group index and FSR are annotated in the plot.

A critical component of this study involved the fabrication and characterization of a series of Mach-Zehnder Interferometers (MZIs) with systematically varied path length differences, denoted as ΔL. The path length differences were designed to span a range from $\Delta L = 50 - 350\,\mu m$, allowing for a comprehensive analysis of the device performance across different optical path delays.

Measured Spectra and Model Fitting:

The primary measurement for each fabricated MZI was the normalized optical transmission spectrum. The measured spectra, which are the core experimental data for this section, are presented visually in Figure 13.

To validate the device design and fabrication, the measured normalized spectra for each interferometer were rigorously compared against the theoretical transfer function of a standard MZI. This fitting procedure utilized the established MZI transmission equation (Eq. 1) that accounts for the path length difference ΔL and the effective index n_{eff}. For each MZI, the measured spectrum was fitted to this model by treating the effective index (n_{eff}) as the sole adjustable parameter.

Observation of Agreement: A key finding of this characterization is the **excellent agreement** observed between the measured spectral data and the predicted theoretical transfer function. This strong correlation validates both the design methodology and the structural integrity of the fabricated devices.

Wavelength-Dependent Loss (WDL): While the overall agreement is high, a closer inspection of the data reveals a subtle but consistent phenomenon: the measured transmission at longer wavelengths shows a discernible dip or reduction compared to the transmission observed at shorter wavelengths. This behavior is consistent across all characterized interferometers and is attributed to **wavelength-dependent loss (WDL)** within the photonic integrated circuit. This loss mechanism must be accounted for in future, more complex integrated optical systems.

Extracted Performance Parameters

The successful fitting of the experimental spectra to the theoretical model allowed for the precise extraction of several critical optical parameters for each MZI: the effective index, the group index, and the Free Spectral Range (FSR).

1. Effective Index (n_{eff}):

The effective index, a fundamental parameter characterizing the light propagation in the waveguide, was directly extracted from the fitting process. The extracted values for n_{eff} were found to range from 2.4472 to 2.452 across the entire set of MZIs.

2. Group Index (n_g):

The group index, which determines the group velocity of light and is critical for dispersion analysis, was derived from the effective index data. The measured group index values ranged from 4.2037 to 4.2104.

Implication for Fabrication Quality:

A highly significant observation is the **narrowness of the measured ranges** for both the effective index and the group index . These narrow ranges are a direct measure of the process variability across the fabricated devices. Crucially, these ranges are **significantly narrower** than the index variations predicted by the initial corner analysis (a worst-case design simulation considering manufacturing tolerances). This superior consistency is a strong indicator of **exceptional fabrication quality** and highly controlled process stability, exceeding the initial predictions for the integrated photonic platform.

Summary:

This report explored the design, theoretical underpinnings, and experimental performance of the Mach-Zehnder Interferometer (MZI) on a silicon photonics platform. The MZI is a fundamental component for integrated optics, and its periodic spectral response is characterized by the Free Spectral Range (FSR), which is inversely proportional to the path length imbalance (ΔL) and the group index.

Modeling and simulation were conducted for a standard 220 nm x 500 nm Silicon-on-Insulator (SOI) waveguide. Simulations confirmed that the FSR decreases as the path length difference increases, with lengths ranging from 50 to 300 μm. Corner analysis and Monte-Carlo simulations were performed to account for fabrication errors, predicting a range of variability in the effective index, group index, and FSR.

Experimental results were based on nine fabricated MZI devices with path length differences ranging from 50 to 350 μm.

- Normalization and Dual-Output: A loopback spectrum was used to normalize the interferometric response. MZI 1 (ΔL = 200 μm) was designed with two output ports, confirming that the normalized spectra from these ports are out-of-phase, as expected from the conservation of energy.
- Repeatability: A study using three identical MZIs (ΔL = 50 μm) showed excellent repeatability and near-perfect agreement with the theoretical transfer function (Eq. 1).
- Path Length Variation: MZIs with different path lengths (ΔL = 50 - 350 μm) also demonstrated excellent agreement with the predicted transfer function.

In addition, the experimentally extracted effective index (2.4472 - 2.452) and group index (4.2037 - 4.2104) ranges were significantly narrower than the variability predicted by the corner analysis, indicating superior fabrication quality.

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